

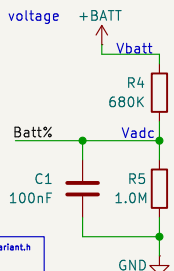
Battery voltage divider for measuring voltage

R4 = 470
R5 = 680

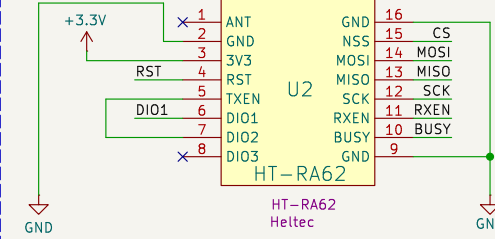
Voltage Division Equation
$V_{adc} = V_{batt} * (R5 / (R4 + R5))$
Voltage $V_{batt} \rightarrow V_{adc}$ ratio
$ADC_Division = V_{adc} / V_{batt} = (R5 / (R4 + R5))$
$R5 / (R4 + R5) = 0.591304347826087$
Voltage $V_{adc} \rightarrow V_{batt}$ ratio
$ADC_Multiplier = V_{adc} / V_{batt} = (R4 + R5) / R5$
$(R4 + R5) / R5 = 1.6913764705882353$

C1 used for getting more stable and accurate readings

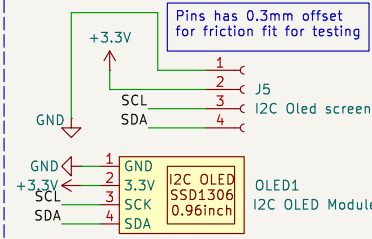
```
firmware/variants/nrf52840/diy/nrf52_micro_diy_tco/variant.h
#define VBAT_DIVIDER_COMP (1.73)
#define ADC_MULTIPLIER VBAT_DIVIDER_COMP
```



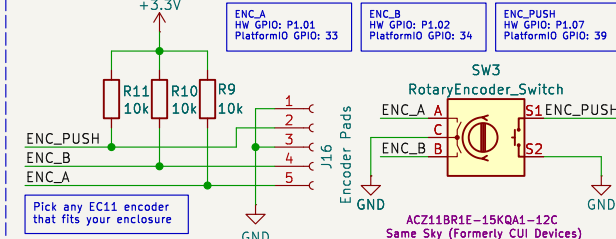
HT-RA62 / RA-01SH



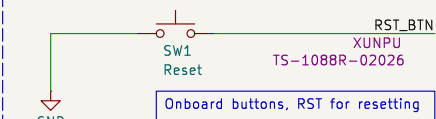
Pins for connecting an I2C OLED screen



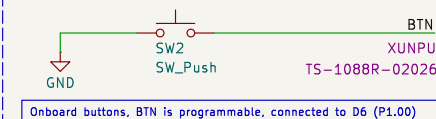
Pads for connecting encoder



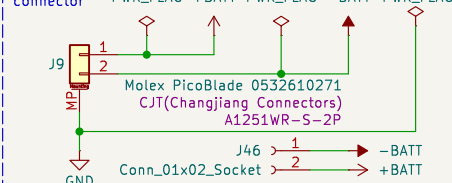
Onboard buttons RST



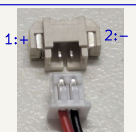
Onboard buttons BTN



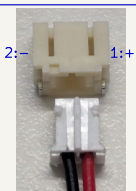
Battery connector



Molex PicoBlade 2-Pin
- MPC: Molex, MPN: 0532610271
- MPC: CJT (Changjiang Connectors), MPN: A1251WR-S-2P
- Pitch: 1.25mm
- Used By: HelTec LoRa 32
- This polarity is also common for lipos in Aliexpress



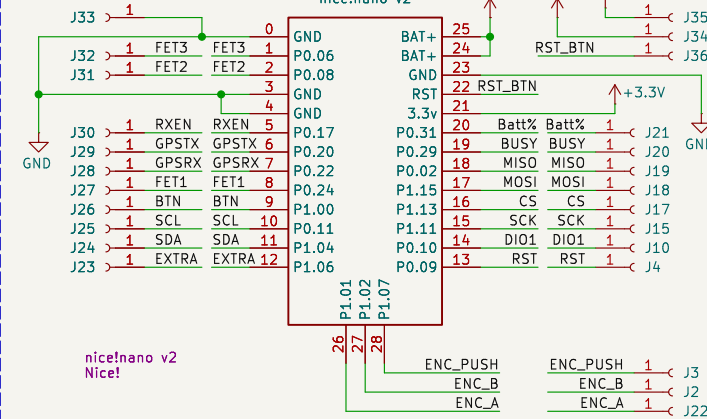
JST PH 2-Pin
- MPN: S2B-PH-SM4-TB
- Pitch: 2.00mm
- Used By: RAK19007 WisBlock
- This polarity is also common for lipos in Aliexpress



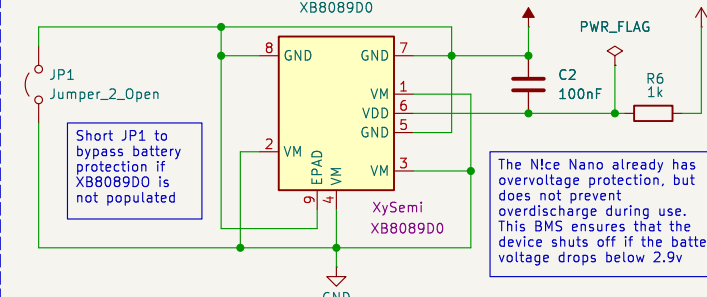
EXTRA GPIO



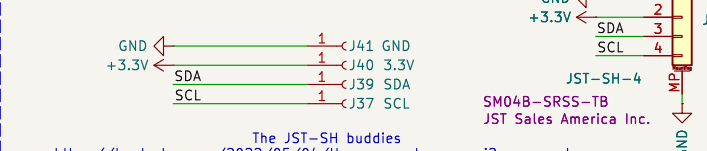
Nice!Nano nrf52840 microcontroller (or promicro)



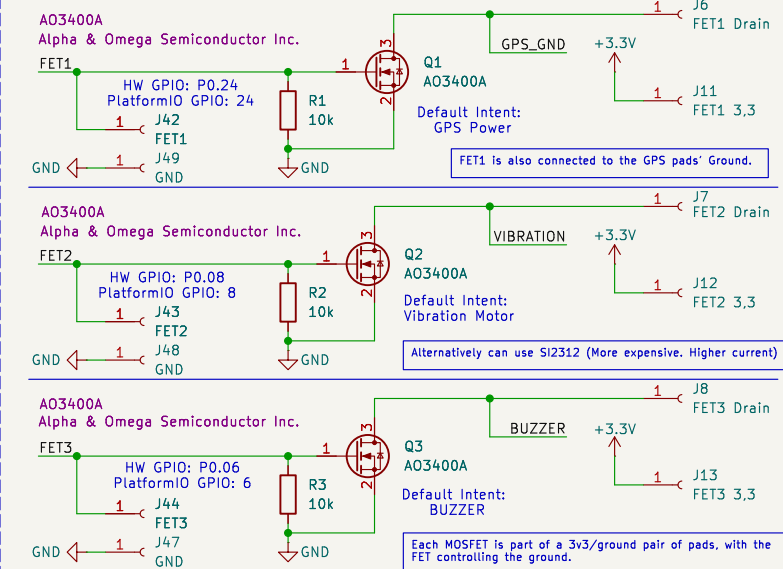
BMS for overdischarge protection



I2C Port (For extra sensors etc...)



Mosfets used for switching high current external peripheral



GPS Module
+3.3V
GPS_GND
GPSRX
GPSTX
GPS Pins
e.g. Buy either ATGM336H or GY-GPS6MV2 modules

Vibration Motor
+3.3V
ERM Motor
VIBRATION
316040001
Seed Technology Co., Ltd
e.g. Can buy in bulk very cheaply from other sources

Buzzer
+3.3V
Buzzer 5020
BZ1
HYDZ
HYG-5020A
Mostly found in LCSC

Development Source: https://github.com/mofosyne/roTec_pcb
Fork Of FakeTec (https://github.com/gargomoma/fakeTec_pcb/)

Sheet: /
File: fakeTecv5.kicad_sch

Title: roTec Rev5

Size: A4
KiCad E.D.A. 9.0.4

Date: 2025-12-25

Rev: 5
Id: 1/1